ModelIt!

Student Activity Pack

# Why Does It Feel Colder When It's Windy?

Modeling Wind Chill and Convective Heat Loss

Grade 06L2 | NGSS MS-PS1-4, MS-PS3-4 | Cause and Effect

STUDENT NAME _____ DATE _____



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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever noticed that a windy day feels much colder than a calm day at the same temperature? Why do you think that happens?**

2. **When you blow on hot soup, why does it cool down faster?**

3. **Does wind actually change the temperature of the air? Why or why not?**

4. **Draw what you think happens to the air around your skin on a windy day versus a calm day.**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Wind Speed | Air Temperature | Convective Heat Loss | Skin Temperature | Perceived Cold

| EXTERNAL | INTERNAL |
|---|---|
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Calm Cold Day

Wind Speed at 0, Air Temperature at 30°F

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Windy Cold Day

Wind Speed at 30 mph, Air Temperature at 30°F

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Wind Chill Danger

Lock Air Temperature at 40°F, increase Wind Speed from 0 to 60 mph

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Humidity Level** — The amount of moisture in the air. Higher humidity can increase heat loss through evaporative cooling, especially on windy days

**Clothing Insulation** — How effectively clothing traps warm air near the body and blocks wind from stripping the thermal boundary layer

**Physical Activity** — The rate at which the body generates internal heat through metabolism and muscle movement, counteracting heat loss

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**If the air temperature is the same on a calm day and a windy day, why does the windy day feel so much colder?** Use evidence from your simulation to support your answers.

**1 Why does the wind chill index warn about frostbite danger even when the actual temperature is above freezing?**

**2 How is the thermal boundary layer similar to wearing a thin invisible jacket?**

**3 Why does wind chill have a diminishing effect at very high wind speeds?**

**4 How could understanding convective heat loss help you dress more effectively for winter?**

**5 Why is wind chill only relevant to warm-blooded organisms and not to objects like cars or rocks?**

# 8 STEM Challenge: Design a Wind Chill Shield

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## YOUR ENGINEERING MISSION

Design a wearable face shield that minimizes convective heat loss while still allowing the wearer to breathe and see clearly.

### THE SCENARIO

A winter sports equipment company needs a new face protection system for skiers and snowboarders who experience extreme wind chill at high speeds. Your engineering team must design a prototype that reduces wind chill effects on exposed skin by at least 50% without fogging up or restricting breathing.

### Engineering Design Thinking

1. WHAT SHAPE WILL BEST DEFLECT WIND WHILE ALLOWING COMFORTABLE BREATHING?
2. WHAT MATERIALS WILL PROVIDE WIND BLOCKING WITHOUT TRAPPING MOISTURE?
3. HOW WILL YOU MEASURE THE TEMPERATURE DIFFERENCE BETWEEN SHIELDED AND UNSHIELDED SKIN?
4. HOW WILL YOU TEST YOUR PROTOTYPE AT DIFFERENT SIMULATED WIND SPEEDS?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Meteorologists and Atmospheric Scientists study weather patterns including wind chill to issue safety warnings that protect millions of people. They earn $55,000-$105,000/year.